EU SAE Application Package

DOWNLOAD AND VERIFICATION INSTRUCTIONS

Version 5.2.0-rc.6-wizard.5.14

Release candidate. This package is a release candidate for review and testing. Whether to publish estimates as official statistics is for each national statistical office to decide. It is published as a GitHub pre-release for review by the country teams.

1. Obtain the correct archive

Download **EU_SAE_520_w5n.zip** and **SHA256SUMS.txt** from Assets of the pre-release on the project Releases page. Its top-level folder is **EU_SAE_520_w5n**. Do not use the automatic "Source code" archives and do not ZIP a working copy of the repository.

2. Verify the package

Open **VERSION** and confirm 5.2.0-rc.6; open **WIZARD_VERSION** and confirm 5.2.0-rc.6-wizard.5.14. From the extracted package root, run **Rscript scripts/verify_release.R** to check the full file set and SHA-256 hashes. Compare the ZIP's SHA-256 with the published SHA256SUMS.txt (Windows: **Get-FileHash <zip> -Algorithm SHA256**; macOS: **shasum -a 256 <zip>**). Stop if a listed file is missing or changed, or if an unexpected file is present.

3. Check the expected contents

The clean package contains the application, R helpers, scripts, tests, output placeholders, instructions, guidance, licence/notice files, and the bundled **Data/simulated/** and **Data/Spain/** examples. Internal notes, .git, app_runs, user-created Data folders, local libraries, credentials, generated outputs, and local diagnostics must not ship. The Spain survey and auxiliary files retain their GPL-2 terms. The boundary uses documented IGN/CNIG CartoBase ANE under CC BY 4.0; see **Data/Spain/README.md** and **THIRD_PARTY_NOTICES.md**. The Spain survey derivation script remains missing.

4. Install and start

Install **R 4.2 or later**. Use the newest R version approved and available in your organization; the absolute latest R release is not required. On a managed computer, keep the approved R installation unless IT directs an upgrade. On first launch the launcher runs **install_packages.R** and installs any missing packages, so keep internet access for that run (you can also run **Rscript install_packages.R** yourself).

HTML and Word reports need Pandoc: the launcher reuses one already installed (RStudio Desktop, Quarto, Positron or standalone Pandoc) and otherwise downloads a pinned, checksum-verified copy into your user profile on first run (26-42 MB, no administrator rights). Without it the analysis still runs and the report is skipped. On Windows, open **Start_Here** and double-click a .bat launcher. On macOS, double-click **Start_Wizard.command** or **Start_Dashboard.command**; if macOS blocks it, open System Settings > Privacy & Security and choose **Open Anyway** (macOS 15 or later), or right-click the launcher and choose **Open** (macOS 14 or earlier). On Linux, run **bash Start_Here/Start_Wizard.command** or the Dashboard equivalent. Keep the terminal window open. See **Start_Here/README.md** for Mac Gatekeeper and executable-permission help.

5. Follow the wizard flow (or dashboard equivalent)

The wizard presents the same inputs as six steps: 1 Data, 2 Mapping, 3 Indicator, 4 Models, 5 AI Assistant (optional consent, key, and language), and 6 Review & Run. In Step 3, the poverty line comes from a survey column or from one numeric value per analysis year (**Poverty line source**). Anchored poverty lines can be entered either way: one value per survey year when a single line applies, or a column when lines vary across records; the app does not calculate them from a CPI. On the final step, use **1. Check Data Readiness** before **2. Run Analysis**. For each auxiliary covariate, Data Readiness reports its correlation with the target indicator with a p-value and significance codes, by year and pooled, most significant first; use it for screening only. Its tables are saved in **outputs/tables/** and kept with the run. The analysis runs in a separate R process: the page stays responsive, the log updates live, and **Stop** ends the run (files written so far stay in outputs/; nothing is archived). The breadcrumb is clickable: green means complete, amber means an outstanding item, and blue means the current step. Navigation is soft. **Run Analysis** stays disabled until the Step 1 data requirements are met, and it starts only after Check Data Readiness passes without an ERROR; check again after changing any input. **Load Last Setup** is available from every step; API keys and transfer consent are never persisted. The classic dashboard exposes the same controls in one sidebar. For mean welfare, Step 3 asks which prices the survey welfare is in (current prices converted with a price index; current prices left as they are, which is the default, so changes include inflation; or constant prices already); the index is typed in or read from an optional CPI file chosen in Step 1, with its organisation, the price base year, and whether incomes refer to the survey year or the calendar year before. Survey, auxiliary, population and target files are keyed by the analysis (survey) year; CPI values by income year and the price base year (for annual CPI formats, also every year in between); geometry by domain ID only. When incomes refer to the calendar year before, the CPI for that earlier year is applied to the survey year. Welfare and poverty lines must share one price basis; uploaded mean-welfare targets must be in the prices of the survey welfare, because the app converts them with the same CPI.

Prepare and run an analysis

INPUTS, MODEL OPTIONS, REPRODUCIBILITY, AND OPTIONAL AI

6. Select approved local inputs

For learning, use either bundled example. **Data/simulated/** contains fully synthetic survey, auxiliary, and geometry files plus a seeded generator. **Data/Spain/** contains the guidance example with mapping, attribution, and licence details. Neither example may feed a real estimate. For actual analyses, browse to approved household survey microdata, domain-level auxiliary covariates, geometry, and optional population or benchmark-target files. The survey file has one row per household with the household weight, household size, and per-person welfare (per capita or equivalised, not the household total); the app weights each household by weight x household size. For a person-level file, use a household-size column of 1s. The auxiliary file has one row per domain and year. Confirm provenance, permitted use, sensitivity classification, area codes, and year coverage before running. Results remain under **outputs/**; see **Data/README.md** for saved-setup compatibility.

7. Apply the revised variance rule

For UFH and MFH, **sm_out** replaces a direct sampling variance with its smoothed value only when the direct value is missing/non-finite or strictly below 0.001. A direct variance equal to 0.001 is retained. **sm_all** replaces all direct variances; **direct** retains direct values subject to documented safety backfills. For UFH the choice is offered only with no transformation: under arcsin the app fills only missing or zero variances, and under log it applies **sm_out** to the log-scale variances.

8. Choose the benchmarking level explicitly

After selecting Apply benchmarking, choose National or Grouped. National uses one countrywide group and constrains population-weighted domain estimates to the direct national target. Grouped requires a higher-level survey variable, such as region or NUTS2, chosen in Step 4 under **Grouped benchmark variable**, and applies the constraint within each group. An external Benchmark Target Database remains optional; its targets are treated as fixed. It needs a year column and a target column (for example **target** or **benchmark**), plus a group column for grouped targets. For mean welfare fitted in logs, the full benchmark operation is repeated on the currency scale inside every bootstrap draw. Survey-derived targets are recomputed; uploaded targets stay fixed. The resulting MSE concerns the final benchmarked estimator.

9. Record reproducibility and inference controls

The seed defaults to 123. **Bootstrap replicates** controls UFH, MFH and benchmarking (default 500; minimum 50). Smaller counts are for quick checks; inspect Monte Carlo precision. Covariance-boundary MFH fits retain point estimates, but change intervals and significance are provisional: more draws alone do not resolve undercoverage. The principal significance flag uses the pointwise raw test; covariance-boundary results are visibly marked provisional. At alpha = 0.05, significance matches whether the pointwise 95% confidence interval excludes zero. Benjamini-Hochberg fields remain sensitivity outputs and should support any statement about how many domains changed.

10. Understand the model selection rule

MFH2 is the default MFH model. MFH1 has independent year effects, MFH2 has AR(1) year effects, and MFH3 has AR(1) year effects with year-specific variances. Selecting MFH3 starts the Molina-Romero sequence: an estimation error or non-converged fit moves automatically to MFH2, with the two cases logged separately. A converged MFH3 fit is retained only when its random-effect variance-homogeneity test supports heterogeneity after the selected multiplicity adjustment. MFH1 remains a manual sensitivity model. For log mean welfare, both models use the log of the area's direct arithmetic mean. UFH **bc_sm** applies emdi's Slud-Maiti correction with ML; MFH uses **Conditional lognormal** prediction. **none** omits bias correction and may underestimate mean welfare. Bootstrap uncertainty repeats back-transformation and benchmarking on the currency scale. The optional MFH3 sensitivity fit does not change the selected model; the final report records the model used.

11. Interpret covariates and fallbacks correctly

Covariate lists are entered in Step 4 (**UFH covariates** and **MFH covariates** for Year 1 and Year 2). Left blank, all eligible numeric auxiliary columns are candidates for stepwise selection. With LASSO enabled, the lists define the screening candidate pool. With LASSO disabled, they define the fixed specification. Tick **PSU codes are consistent over time** (Step 4, Data assessment) only if the same PSU codes identify the same sampling units in both years. In log runs for mean welfare, zero and negative household welfare remain in the arithmetic direct mean. Every modeled area's mean must be positive. Check Data Readiness reports invalid area means. A valid zero-variance MFH boundary retains point estimates and can be bootstrapped, but its change inference remains provisional; failed REML endpoints cannot support inference.

12. Treat AI as a separate external transfer

AI is off by default. Tick **Enable AI Assistant** and the external-transfer acknowledgement, and confirm that the provider or gateway is institutionally approved, before entering a key. Keys starting sk-ant- go to Anthropic and sk- to OpenAI; an approved gateway is set with the SAE_ANTHROPIC_BASE_URL or SAE_OPENAI_BASE_URL environment variable (see **docs/SECURITY_AND_AI.md**). AI text is generated during Run Analysis when Comparison is selected, so enable it before running. The application does not send household microdata. Prompts include unpublished aggregate estimates, uncertainty measures, diagnostics, counts, and analysis context; geographic identifiers are removed from top-ranked and extreme-change prompt tables. Aggregate unpublished estimates may still be confidential. AI commentary is advisory and requires documented human review before dissemination; the report marks each AI block 'Human review: not recorded', so record the review in your own approval file.

Keep the evidence and complete review

RECORDKEEPING AND PRODUCTION RELEASE

13. Read both reports and keep the evidence

After a successful run, open **outputs/final_report.html** in a browser or **outputs/final_report.docx** in Word. Word text and tables are editable, and figures are embedded images. Save a separate working copy before editing; later runs replace current outputs. Both formats contain the same statistical results and labeled AI text when requested. Keep **outputs/data/** beside the reports for linked Excel workbooks; archived copies are under **app_runs/<timestamp>_<run_label>/outputs/**.

The clean ZIP excludes generated reports. Signed-change figures are under **outputs/figures/change_comparisons/**, with source values in **outputs/data/change_estimate_comparison.xlsx** and **EU_SAE_results.xlsx**. Poverty changes are in percentage points; mean-welfare changes in the maps, significance plots, and paired figures are percentage changes of the mean, and the change-and-RMSE box plots show changes in ln mean welfare, not multiplied by 100 (0.029 is about 2.9%). These figures compare estimated changes, not confidence-interval widths. A tighter spread across domain estimates alone does not establish smaller MSE or a statistically significant variance reduction. All maps use one colour scheme. In the final report, maps of one kind (poverty rates, RMSE, poverty change) share one legend range, so colours can be compared across methods and years; maps on the Spain boundary carry the IGN credit.

If report conversion fails, the HTML remains. Check the run log (it names the Pandoc folder used, or why none was available), xml2, and zip, then regenerate after resolving the warning. Retain the exact ZIP, release_manifest.csv, verification output, VERSION, WIZARD_VERSION, run metadata, input hashes, and review approvals. Passing tests or checksums does not approve the statistical methods, data rights, privacy basis, or official publication.

Before a production release. Complete **docs/RELEASE_CHECKLIST.md**, obtain institutional approval for documentation/media rights and release authority, and independently verify the release manifest.